

1. Prove the following statement for all positive integers n using mathematical induction. [6 pts]

$$\frac{1}{(1)(2)} + \frac{1}{(2)(3)} + \frac{1}{(3)(4)} + \cdots + \frac{1}{(n)(n+1)} = \frac{n}{n+1}$$

2. Expand $\left(2 - \frac{1}{x}\right)^5$ Simplify your expansion so that there is a single coefficient with no choose numbers. [3 pts].

3. Find the coefficient of $x^{10}y^{40}z^{50}$ in the expansion of $(3x - y + 2z)^{100}$. No need to simplify your coefficient (product with choose numbers and/or powers is fine). No justification required. [3 pts]

4. Find the coefficient of x^{49} in the expansion of the product $(2 + x)^{22}(3 + x^2)^{15}$.
Show the work that leads to your answer. Again, no need to simplify. [3 pts].